A Primer on Open Source IoT Middleware *for* the Integration *of* Enterprise Applications

The Internet of Things (IoT) integrates a virtual world of information to the real world of devices through a layered architecture. IoT middleware is an interface between the physical world (hardware layer) of devices with the virtual world (application layer), which is responsible for interacting with devices and information management systems. This article discusses IoT middleware, the characteristics of open source IoT middleware, IoT middleware platform architecture and key open source IoT middleware platforms.

With billions of devices generating trillions of bytes of data, there is a need for heterogeneous IoT device management and application enablement. This requires a revamp of existing architectures. There is a need to identify industry agnostic application middleware to address the complexity of IoT solutions, future changes, the integration of IoT with mobile devices, various types of machinery, equipment and tablets, among other devices.

According to *Statista*, the total installed base of IoT connected devices is projected to be 75.44 billion worldwide by 2025.

Most of the IoT applications are heterogeneous, and domain specific. Deciding on the appropriate IoT middleware for app development is the major challenge faced by developers today. The functionalities provided by different middleware vendors are almost similar but differ mainly in their underlying technologies. Middleware services provided by different IoT vendors include data acquisition, device management, data storage, security and analytics. Selecting the right middleware platform is one of the critical steps in application development.

The important parameters for choosing the right middleware for an IoT application are scalability, availability, the ability to handle huge amounts of data, a high processing speed, flexibility, integration with varied analytical tools, security and cost.